

Assignment. <4.3>	
<Pointers>	
Course Code: CPE007	Program: Computer Engineering
Course Title: Programming logic and design	Date Performed: 9/22/2025
Section: CPE11S1	Date Submitted:9/22/2025
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6. Output 1. What is a pointer in C++? A pointer is a variable that stores the memory address of another variable. 2. How does a pointer differ from a regular variable? A regular variable directly stores a value A pointer stores the address of a variable 3. What operator is used to get the address of a variable? The ampersand (&) operator. 4. What operator is used to access the value stored at a pointer's address? The asterisk (*) operator, or the dereference operator. 5. Why are pointers important in C++? Give two uses. Dynamic memory management using new and delete to allocate and free memory. Efficient array and function handling arrays are naturally handled with pointers, and pointers allow passing large data to functions without making copies.	
7. Supplementary Activity 1. <pre>int x = 42; int *ptr = &x; cout << *ptr;</pre> first create an integer x to stores the value 42, then you create a pointer to connect in to the x Then use cout to print the value x and the output is 42 2. <pre>int a = 5, b = 10; int *p = &a; p = &b; cout << *p;</pre> first this code assign the *p =&a which is 5 then it reassigned to p = &b; then in the cout put *p to assign the pointer to switch to p =&b; the output is 10	

3.

```
int arr[3] = {10, 20, 30};  
int *p = arr;  
cout << *p;
```

first create an arr[3]. 3 because in the {} it has 3 elements, then put a pointer to arr then use cout put *p to get the first element of arr. **The output is 10**

4.

```
int arr[4] = {2, 4, 6, 8};  
int *p = arr;  
p++;  
cout << *p;
```

At first the program will print 2 but you put p++; that means it will increment 1 so the **output is 4**

5.

```
int arr[3] = {5, 15, 25};  
int *p = arr;  
cout << *(p + 2);
```

the cout pointer will add up p + 2 so the output is 25

1.

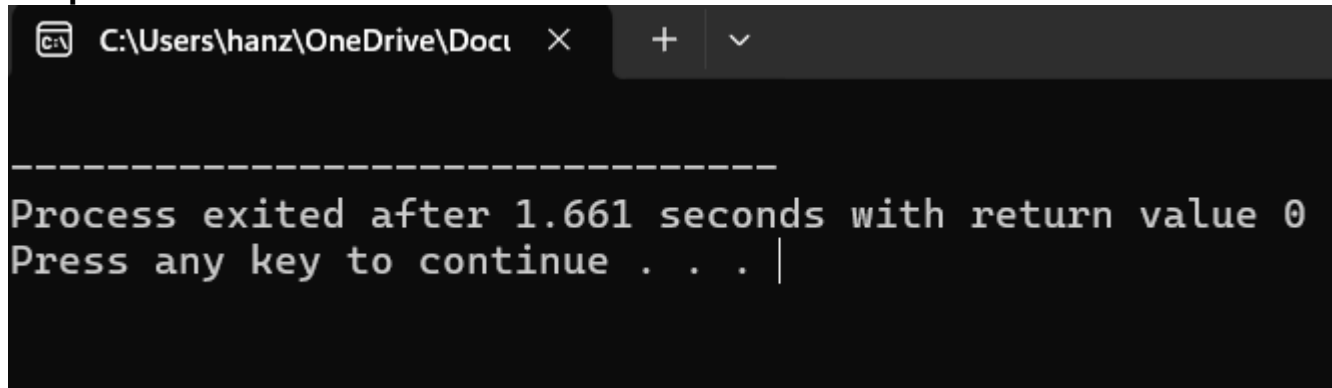
```
int arr[3] = {1, 2, 3};  
int *p = &arr;
```

Code:

pointers.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5
6      int arr[3] = {1,2,3};
7      int *p = &arr[0];
8
9
10     return 0;
11 }
```

Output:



The screenshot shows a Windows command prompt window with the title bar "C:\Users\hanz\OneDrive\Docu". The window contains the following text:

```
-----
Process exited after 1.661 seconds with return value 0
Press any key to continue . . . |
```

Explanation: put [0] to run the code. But no output because theres no cout.

2.

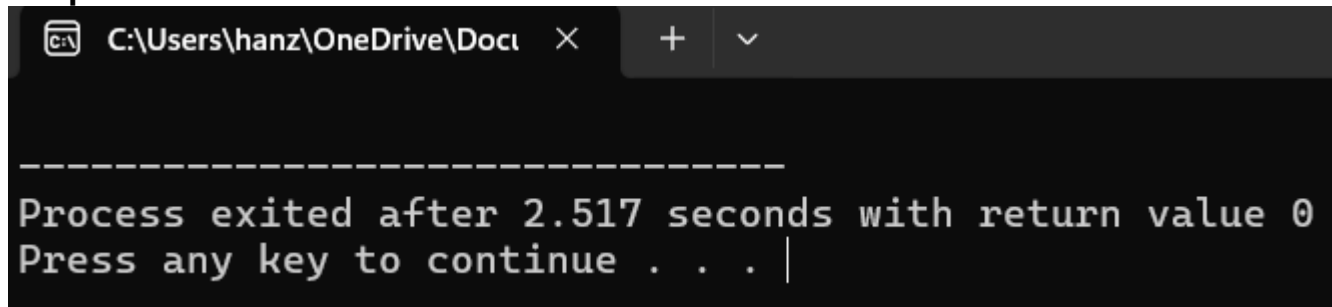
```
int arr[5];
int *p;
p = arr[2];
```

Code:

pointers.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5
6      int arr[5] = {1,2,3,4,5};
7      int *p;
8      p = &arr[2];
9
10     return 0;
11 }
```

Output:



```
-----
Process exited after 2.517 seconds with return value 0
Press any key to continue . . . |
```

Explanation: put 5 numbers in the {} and no output because theres no cout.

3.

```
int arr[4] = {10, 20, 30, 40};
```

```
cout << *arr[2];
```

Code:

pointers.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5
6      int arr[4] = {10,20,30,40};
7      cout<<arr[2];
8
9      return 0;
10 }
```

Output:

```
C:\Users\hanz\OneDrive\Docu  X + v
30
-----
Process exited after 1.691 seconds with return value 0
Press any key to continue . . . |
```

Explanation: I just remove the * in the arr

8. Conclusion

In this topic I focused on pointers, operators and help me to understand how it works, the operators is kind of confusing at first because I don't know where I should put it, but I do some studying that's why I can analyze the code. And at analyzing it really practice my error sighting also I really enjoy when I analyzed a problem so I can see my my knowledge in coding, my reflection here is I can do it when I study smart, you can study on your own and you can upgrade your skills and just do study and will understand the lessons, or for the future you know what you can do.

9. Assessment Rubric

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